

Thus far, we've learned how to compute the 'volume under the surface' $z = f(x, y)$ if its defined over a nice, simple region R . An immediate application of this machinery is the ability to quickly compute 2-dimensional area and a surface's average value.

Given a region R in the xy -plane, it is reasonable to ask for its area. Using single-variable methods, we learned how to accomplish this if R is, say, y -simple:

However, this is really just a double integral in disguise. Namely, consider the surface of height 1: $z = f(x, y) = 1$. Then we see that

$$\iint_R f(x, y) dA = \iint_R 1 dA = \text{Area of } R$$

Example. Find the area of the region R bounded by $y = x$ and $y = x^2$ in the first quadrant.

Example. Find the area of the region R enclosed by the parabola $y = x^2$ and the line $y = x + 2$.

Example. Consider the region R bounded by the lines $y = 3x$, $y = \frac{x}{3}$, and $y = 4 - x$. Sketch this region in the xy -plane and then compute its area using iterated integrals.

Once we have a quick and easy way to compute the area of a region R , we can now talk about the ‘average value’ of a surface $f(x, y)$ defined over R . Generally we know that

$$\text{‘Average’ of } output \text{ values} = \frac{\text{sum of your } output \text{ values}}{\text{number of } input \text{ values}}.$$

Therefore, if we have a surface $f(x, y)$ defined over R ,

$$\textbf{Average value of } f \text{ over } R = \frac{1}{\text{area of } R} \iint_R f(x, y) dA.$$

Example. Find the average value of $f(x, y) = x \cos(xy)$ over the rectangle

$$R: 0 \leq x \leq \pi, \quad 0 \leq y \leq 1.$$

Example. Find the average value of $f(x, y) = xy$ over the quarter disk R given by $x^2 + y^2 \leq 1$ with $0 \leq x \leq 1$ and $0 \leq y \leq 1$.